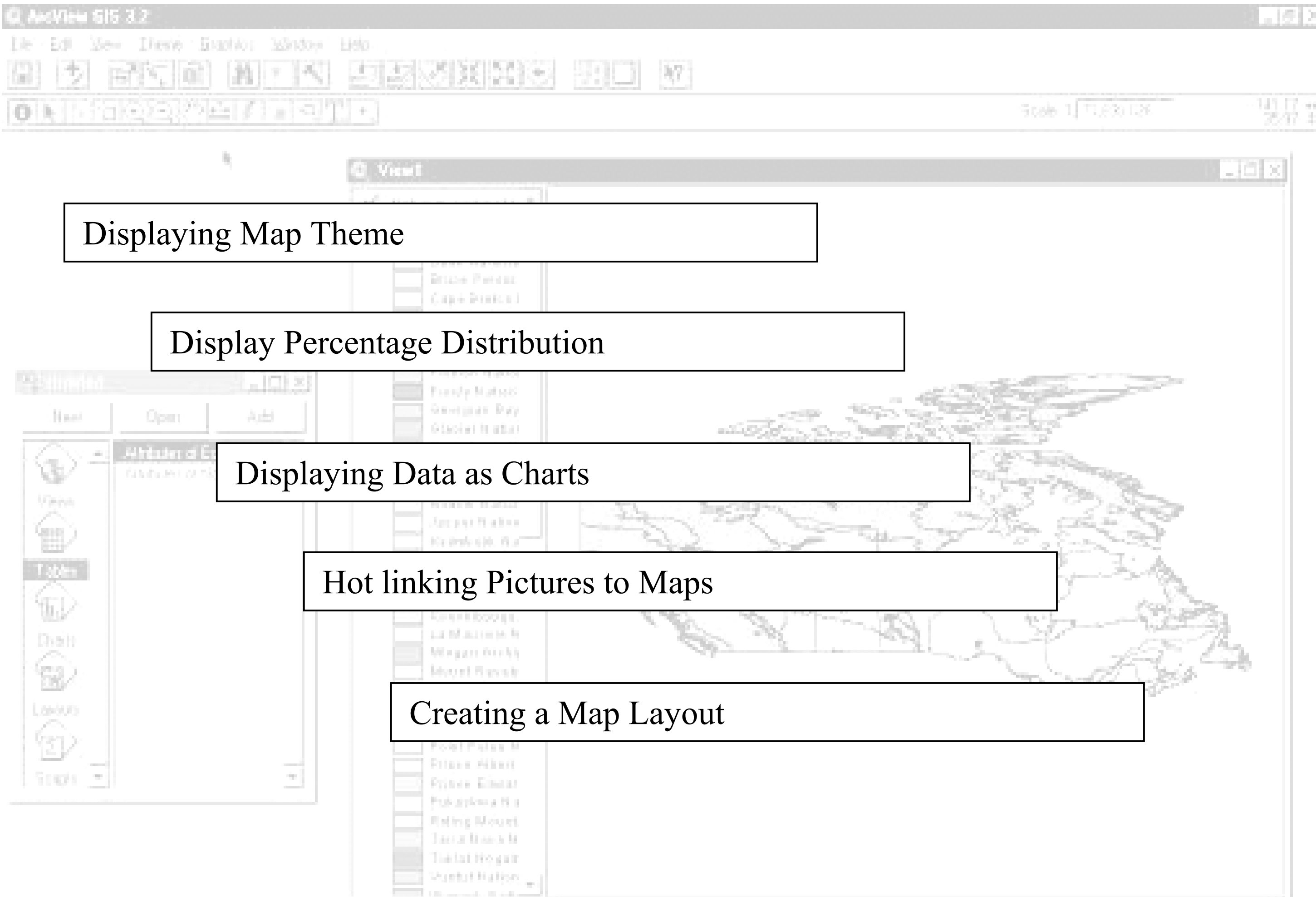
Landforms of Canada



Displaying Map Theme

Display Percentage Distribution

Displaying Data as Charts

Hot linking Pictures to Maps

Creating a Map Layout

ArcView Lesson

GIS Information

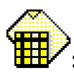
Here's a list and explanation of important features in ArcView:

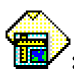
Project : In ArcView, a file for organizing your work. Projects use five types of documents to organize information: views, tables, charts, layouts, and scripts.

View  **:** A component of an ArcView project used for displaying, querying, and analyzing geographic themes.

Table of Contents: All themes in a View are listed to the left of the map. The Table of Contents show the symbols used to draw features in each theme. The check box next to each theme indicates whether or not it is turned on or off in the map (whether it is currently drawn on the map).

Theme: A set of related geographic features, such as streets, parcels, or rivers, and the characteristics (attributes) of those features.

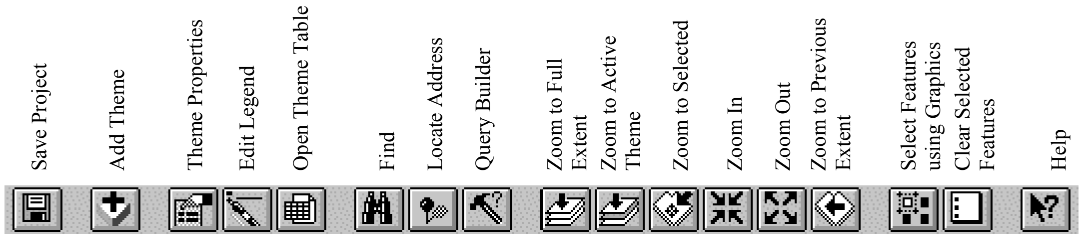
Tables  **:** Information formatted in rows and columns. A component of an ArcView project used for displaying Tabular data.

Layout  **:** The design or arrangement of elements in a digital map display or printed map. A component of an ArcView project used for creating presentation-quality maps.

Hotlink: In ArcView, a way to display data (for example, a file, image, ArcView document, or project) directly from a view, by clicking on a feature.

ArcView Buttons Reference Guide

View - Buttons



View – Tools

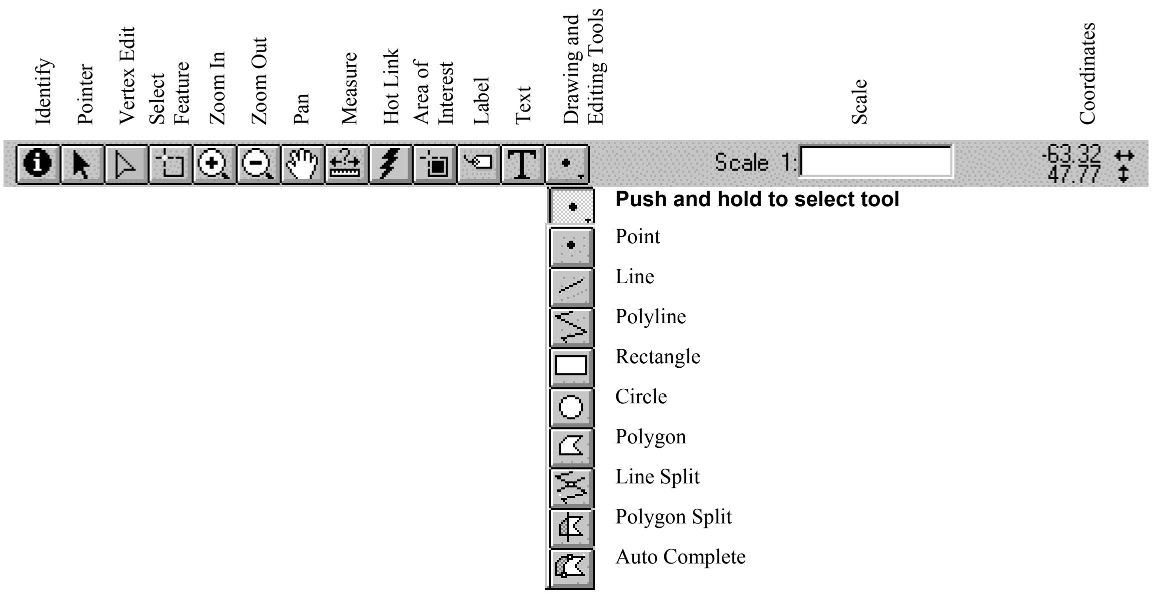


Table - Buttons

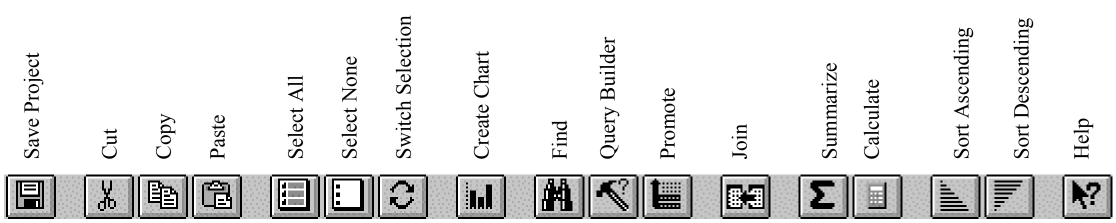


Table - Tools

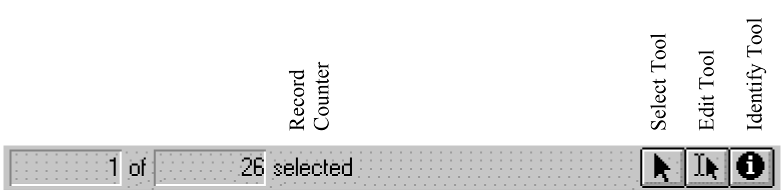


Chart - Buttons

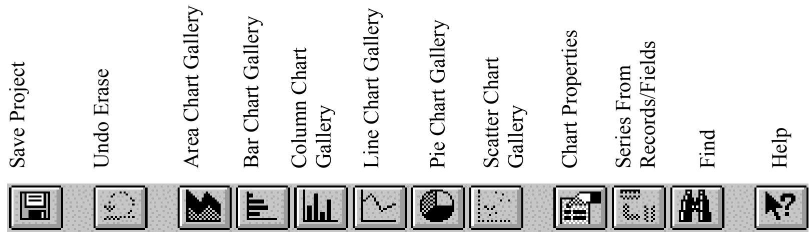
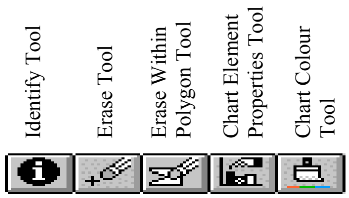
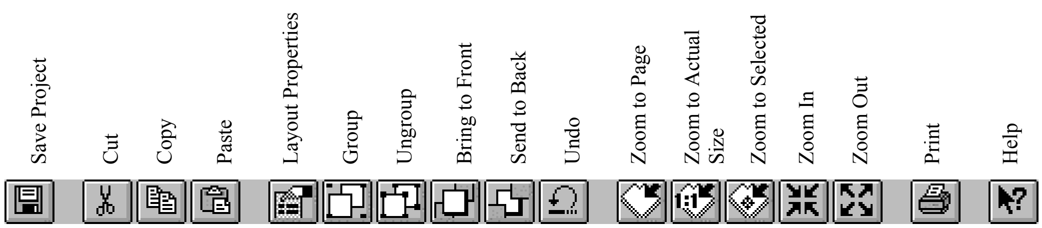


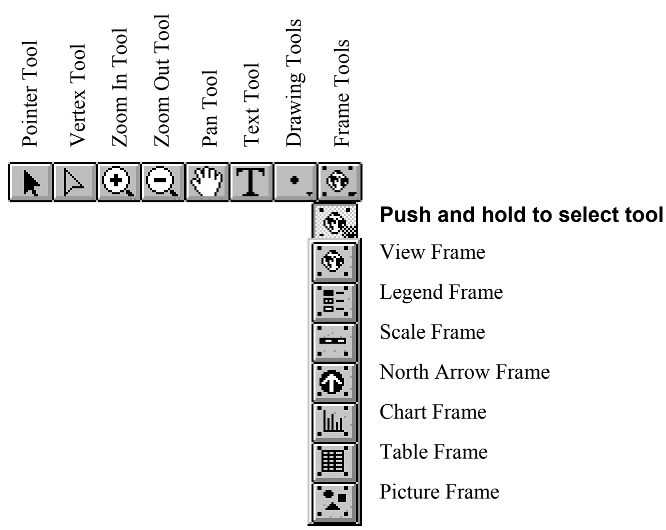
Chart - Tools



Layout - Buttons



Layout - Tools



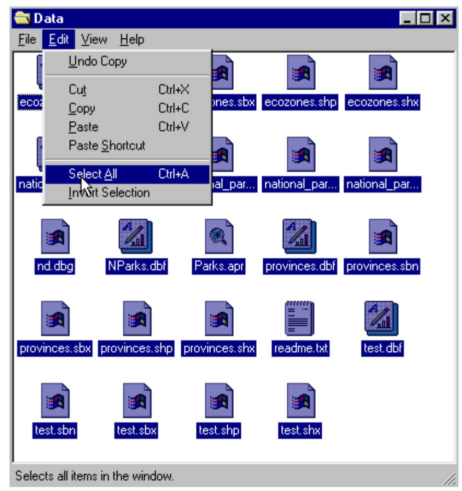
For more Information: http://www.esricanada.com

Getting Started

Before you begin, the lesson data folder must be ***copied*** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

! *(The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).*

To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /.**

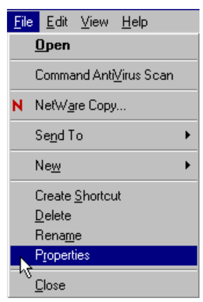


Once the data folder has been ***copied*** to the required destination the files must be made usable by removing the ***read only*** attribute. For windows based systems follow these instructions:

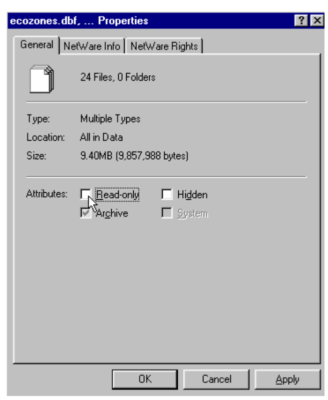
! *(The data downloaded from the Web Site is set as **read only** so that it cannot be changed or altered by accident.)*

1. Open your new data folder
2. Choose **Edit >** ***Select all*** (they will all appear Blue)

3. From the menu choose:
   File > Properties



4. Remove the check mark in the box beside the label ***read only***.



5. Click **apply**, then **OK**.

The data in your new data folder can now be opened, edited and saved using ArcView or other GIS software that can use SHAPE files to carry out the following project.

! Denotes an observation.

? Denotes a question to be answered on the Student Activity Sheet.

STARTING THE PROJECT

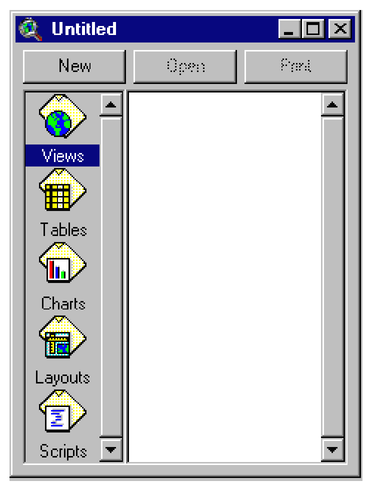
The purpose of this exercise is to practice display functions using data tables, to display percentage distribution and data charts, compare distribution display techniques, to learn to hot link pictures to a map.

STARTING ArcView

Begin by starting up the **ArcView** program. If you are not sure how to do this, ask your teacher.

1. Choose *as a blank project*
2. Click **OK**.

When the program starts, it automatically opens a project window titled **"Untitled."**

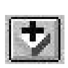


If you like, you can maximize the **ArcView** window to fill the screen.

! *In ArcView, you work within a project. A project is a collection of documents (map views, tables, charts, layouts, etc), document user interfaces, (called DocGUIs) and scripts. Only one project is open at a time.*

3. Now you are ready to add documents to your new project.

4. With the **"Views"** icon selected, click on **New** and maximize the **View1** window.

5. Click the **Add theme button**  and Navigate to the data folder where your data is stored. Hold down your **SHIFT** key and select the following files:

Admin.shp
Ecozones.shp

6. Click **OK**.

This will add these two themes to the table of contents of the new view. They are now available for display and editing.

7. Click and drag to move the themes up or down in the table of content to change the order of display if necessary.

! Note the themes are displayed from the bottom to the top so line and point themes should be on top of polygon files to be visible.

8. **Click** on the square next to the themes you want to display. The theme that appears raised in the table of contents is the current active theme. This can be change by **Clicking** on other themes.

9. Remember to **Save** your project as you go along. Go to the **File** menu and select **Save Project.** Save your work to the appropriate directory (ask your teacher if you are not sure). Give your project a name that will identify your work. (The suffix **apr.** means that you are saving your work as a project that can be opened up in the future)

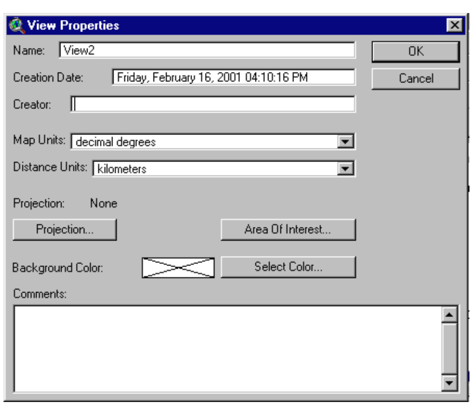
ArcView - Changing the Map Projection

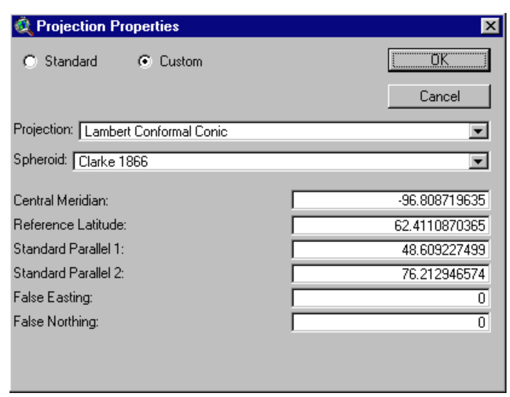
! A ***Map Projection*** defines the relationship between the map coordinates and the geographic coordinates, (latitude and longitude). Our current display shows what a map of Canada looks like with only a geographic projection.

CHANGING PROJECTION

Now we will give our map "projection coordinates" to make it look more like the maps which are familiar to us.

1. From the **View** menu, choose **Properties**.
2. Select **Properties**.
3. Select **Map Units** list. Change the **map units to** *decimal degrees.*
4. Select **Distance Units** list. Change the **distance units to** *kilometers*
5. Click the **Projection** button.
6. Select **Custom**.
7. Select **Lambert Conformal Conic** from the projection list.



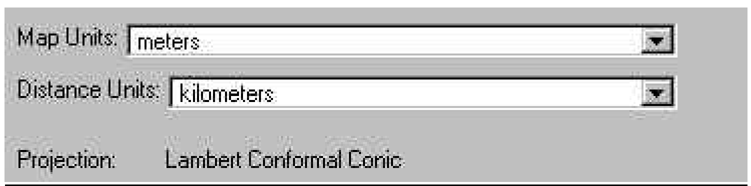


! Take a look at the parameters for this projection. Parameters are the descriptions of how the projection was made.
Note that a spheroid was used. (The earth is a slightly flattened sphere toward the poles).
Conformal implies that all angles are indicated correctly. All angles measured on the earth's surface are given the same values on the map.
Conical means in the shape of a cone. In this projection, the relative positions on a globe are transferred to a plane in the shape of a cone around the globe. See how the longitudinal lines are closer together at the top of the image, that is, they are not parallel.

The ***central meridian***, ***false easting*** and ***false northing*** are used to define the X (easting) and Y (northing) coordinates of the map. The ***reference latitude*** and ***standard parallel 1 and 2*** are used to determine the origin of the projection (i.e. the point from which you are viewing the globe).

8. Click **OK**.

The bottom of the **View Properties** window should look like this:



9. Click **OK**.

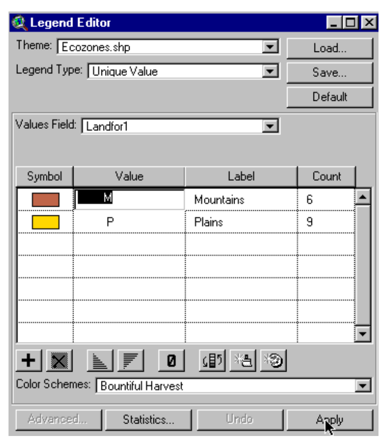
! Note the difference in the map presentation.

? How is it different from the non-referenced map?

ArcView - DISPLAYING MAP THEMES

We will be looking at the four main landforms of each ecozone in Canada.

1. **Double Click *Ecozones*** theme.
2. Open **Legend Type** list, Select **Unique Value**.
3. Open **Values Field** list, Select ***Landfor1***.
4. Edit the legend by clicking ***M*** in the **Label Field**, type ***mountains***.
5. Repeat for ***P***, type ***plains***.
6. Click **Apply**.
7. Close **Legend Editor** window.

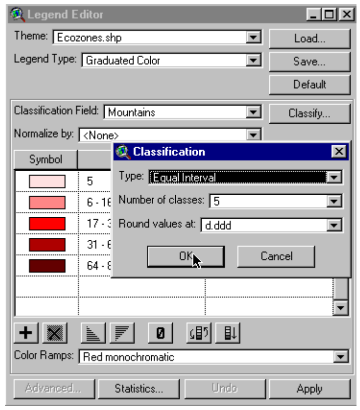


? What can you say about the distribution of the two primary landforms (Mountains and Plains) i.e. where they are located in Canada?

DISPLAY PERCENTAGE DISTRIBUTION

We will display the percentage distribution of the mountain landform.

1. **Double Click *Ecozones*** theme.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.
5. Select ***Mountains***.
6. Click **classify** button.
7. Open **Type** list.
8. Select **Equal Intervals**.
9. Click **OK**.
10. Click **Apply**.
11. Close **Legend Editor** window.



? **What can you say about the distribution of mountains?**

? **Are there any ecozones not represented?**

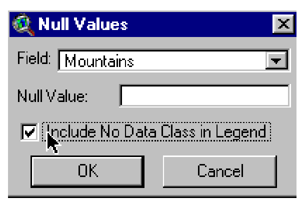
? **Why?**

! To have a completed map, we will put a no data item in our legend for those ecozones that do not have a particular landform.

12. **Double Click *Ecozones*** theme.

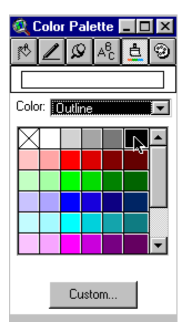
13. In the **Legend Editor** window, Click on the  **Null Value** button.

14. Check **Include No Data Class In Legend** box
15. Click **OK**.



16. **Double Click** no data legend item.

17. Click **The Colour Palette** button.
18. Select a new colour for the **no data** legend item.
19. Open **The Colour** list, **Select Outline**.
20. Select **black** for outline colour.
21. Close **Colour Palette** window.
22. Click **Apply**.
23. Close **Legend Editor** window.

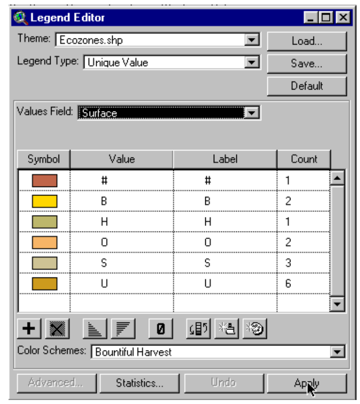


! Using the same process that was used to display mountains, display the other landforms of plains, hills, plateau and valley one at a time. Explain the distribution patterns you see.

? Explain the distribution patterns you see for Plains, Hills, Plateau and Valleys.

Now that we know where the major landforms are located, let's explore to see what the surface forms are.

First we'll display the primary surface forms or those, which occur most frequently.



1. **Double Click *Ecozones*** theme.
2. Open **Legend Type** list, Select **Unique Value**.
3. Open **Values Field** list, Select ***Surface***.
4. Edit the **label** values to represent the following.

B - BOG
O - ROLLING
H - HUMMOCKY
- NOT APPLICABLE
S - STEEP
U - UNDULATING

5. Click **Apply**.
6. Close **Legend Editor** window.

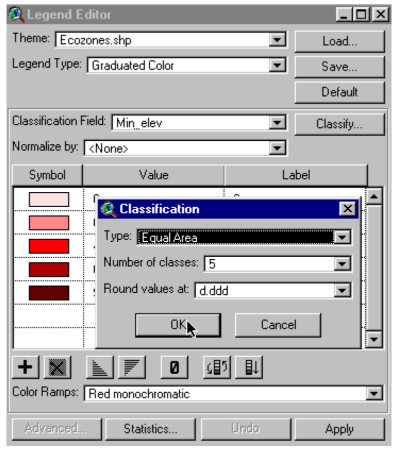
7. Use the **Identify** button  to identify a representative ecozone for each of the primary surface forms.

Now let's look at the minimum and maximum elevation characteristics of the Ecozones.

Elevation or height above mean sea level is an important measure of landform. Elevation for an Ecozone can be quite varied and therefore difficult to present on a map. Statistical values of elevation are often used to simplify presentations. Four common values that are used are;

1. The **Minimum Elevation** found in the Ecozone. This actually may be **Sea Level**.
2. The **Maximum Elevation** found in the Ecozone.
3. The **Mean** or Average Elevation found in the Ecozone.
4. The **Elevation Difference**, which is difference between the Minimum and Maximum Elevation.

1. **Double Click *Ecozones*** theme.
2. Open **Legend Type** list.
3. Select **Graduated Colour**.
4. Open **Classification Field** list.
5. Select ***min_elev*.**
6. Click **Classify** button.
7. Open **Type** list.
8. Select **Equal Intervals.**
9. Click **O.K.**
10. Click **Apply**.
11. Close **Legend Editor** Window.



Repeat the process to display ***max_elev.*** Maximum elevation.

? Describe the different patterns displayed when illustrating the Minimum Elevations of Canada. Explain why.

? Now compare with the Maximum Elevations of Canada. Is this the same pattern as for Minimum Elevations? Explain.

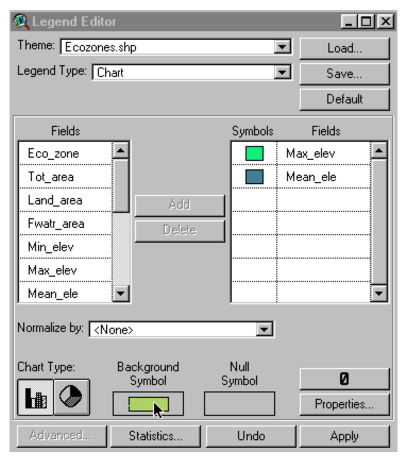
Repeat the process for the ***mean_ele*** mean elevation and the ***elevdiff.*** Elevation difference.

? How do mean elevation and elevation difference compare to the previous patterns?

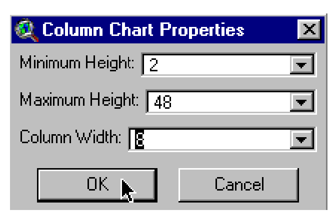
ArcView - DISPLAY DATA AS CHARTS

We will now display the maximum and mean elevations as charts for each ecozone.

1. **Double Click *Ecozone*** theme.
2. Open **Legend Type** list.
3. Select **Chart**.
4. In the **Fields List**, Select ***Max_elev.***
5. Click **Add**.
6. **Repeat** for ***Mean_ele***.
7. For **Chart Type** Click **Bar Chart**.
8. **Double Click Background Symbol** and Select a **Background Symbol** that does not conflict with your legend colours.



9. In the **Legend Editor** window, Click **Properties** to alter the look of the charts.
10. Click **OK.**
11. Click **Apply**.
12. Close **Legend Editor** window.



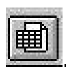
? By comparing the charts, which ecozones show the largest elevation difference?

? Which ecozones have very little elevation difference?

? By regrouping the surface forms, which ecozones
- **-have the most wetland?**
- **-would be the easiest to traverse?**
- **-would have the steepest inclines?**

ArcView - HOT LINKING PICTURES TO A MAP

1. **Double click *ecozones*** theme.
2. Reset **Legend type** back to **Single symbol**.
3. Click **Apply**.
4. Close **Legend Editor** window.

5. Click **Open theme table** button .

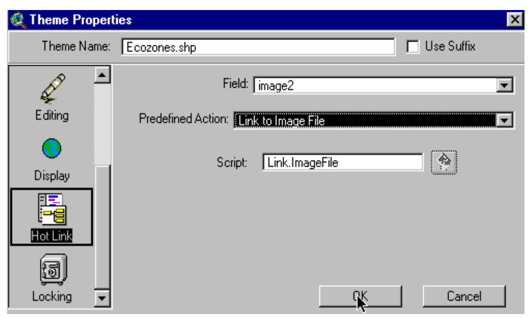
6. Scroll a cross the table to the ***Images 1*** field. Note the values in this field all end as a **.gif** file. These are pictures representing the unique characteristics of the Ecozones of Canada. We are going to link these images to the Arc View project.

7. With the ***ecozones*** theme active, select **Theme** in the **main menu**. Select **Properties**.

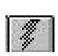
8. Scroll down the icon list on the left until you come to the **Hotlinks**.

9. Click on it.

10. Open **Field** list.
11. Select ***image 1***.
12. Open **Predefined action** list
13. Select **Link to Image File**.
14. Click **OK**.



Now we'll test our hot links.

! Note that the **hot link** button  is now active.

15. Click on it.
16. Move the cursor to an ecozone and click.

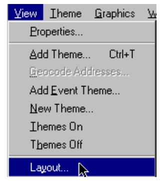
? What happens when you select an ecozone that you linked to an image?

Now create a layout for printing. Use examples from this exercise to show an aspect of Canada's landforms.

ArcView - Map Layout for printing

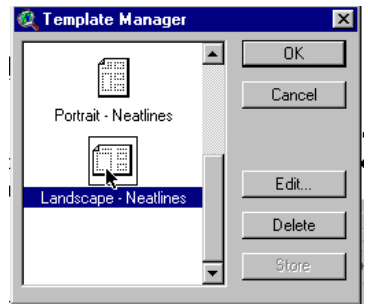
Once you have a map classified and displayed to your satisfaction, you should prepare a map layout that can be used for printing. The purpose of this lesson is to set up your map on a page and annotate it for presentation.

SET UP MAP



1. Click Menu **view.**
2. Click **Layout.**

3. Select the **layout template** you want to use (One of the landscape templates is suggested) .



4. Click **O.K**.
5. Maximize the **Layout** screen.

EDIT TITLE

1. **Double Click** on the title ***View1***.
2. **Edit** the title to reflect the purpose of the map, e.g., forestry, and population.
3. Click **O.K.**
4. **Resize** the title using the corner handles and move it into a position that is visually balanced.
5. **Click and Drag** to reposition the legend if necessary.

EDIT NORTH ARROW

1. **Double Click** on the *north arrow* and select a different presentation. Ensure that that the arrow is pointing in the representative north direction of the map.
2. **Click and Drag** to reposition or rotate (using rotation function) as needed.

EDIT SCALE BAR

1. **Double Click** the *scale bar.*
2. **Choose** a new style of scale bar. The scale bar has default properties so you will not be able to change everything to the way you want.
3. **Change** the units to ***kms.***
4. Find a combination of style, scale and divisions that is visually pleasing
5. **Click and Drag** to position to an appropriate location.

ADD LABELS, TEXT, OTHER INFORMATION

1. **Click** on the **Text** tool.
2. **Click** at the position on your layout where you want the text to appear.
3. In the **Text Properties** box, enter your text, e.g., “produced by (your name, date)”.
4. **Select** desired formatting information such as **alignment, spacing and rotation.**
5. **Click O.K**. when it appears, as you want it.
6. **Click** on **Pointer** tool.
7. **Select** the new text

8. **Click once** to change location or size, Click twice to change content.

! Experiment with all of the components of the layout until you have the look you want.

CHANGE THE NAME OF THE LAYOUT

You only have to do this if you want to save your project on the computer or on a disk.

1. **Click** menu **Layout.**
2. **Select Properties.**
3. **Change** the **"name".** Click **O.K.**

ARRANGING YOUR LAYOUT

! Experiment with the layout buttons (use the tools page provided at the beginning of this lesson)

- to **group** (tie pieces together so they move as one)
- to **ungroup** (break apart the ties so that individual pieces move)
- **bring to front** (change order that pieces were laid down, e.g., lines on top of polygons)
- **send to back** (change order that pieces were laid down, e.g., polygons behind lines)
- **zoom** (make image bigger or smaller)
- **neat line** (border around layout)

PRINT MAP LAYOUT

1. **Click Print** button.
2. **Click Setup.**
3. **Select** your printer properties as outlined by your teacher.
4. **Click O.K.** to print.

! **Note** that the map can also be saved to a printer file.

? **Produce a layout for printing your map. Include your layout with this exercise sheet for evaluation. What is the purpose of your map?**